Layer 7: SAM Chemistry

Mixed Probe-Thiol/MCH Monolayer: Deposition, Stability, and Ca^{2+} Artefact

Eton iGEM 2026, Dry Lab | Layer 7 of 17-Layer Biosensor Pipeline | `sam_loading.py`, `sam_stability.py`, `calcium_artefact.py`

Layer 7 characterises the self-assembled monolayer (SAM) that immobilises the aptamer probe on the gold electrode surface. Its primary output is the probe surface density Γ_p, which passes to Layer 8 (steric accessibility) and Layer 9 (surface kinetics). Secondary outputs are bounds on SAM degradation over the 60-minute assay window and a quantitative account of the Ca^{2+} artefact and its ratiometric cancellation.

Every parameter value must be confirmed from a paper physically present in `papers/layer_07/`; values without a paper source are explicitly flagged throughout.

I. Mixed SAM Architecture

Molecular components

The sensor electrode carries a two-component thiol monolayer. The probe-thiol consists of a peptide substrate bearing a methylene-blue (MB) redox label, tethered to gold via a C_6 alkyl spacer (HS–$(CH_2)_6$–peptide–MB). MMP-9 (or HNE on the secondary channel) cleaves the peptide, releasing the MB-bearing fragment into solution and producing a signal-off decrease in SWV peak current proportional to enzymatic activity (Layer 0, Section V).

The diluent is 6-mercapto-1-hexanol (MCH, HS–$(CH_2)_6$–OH). MCH fills gold sites not occupied by the probe-thiol, presenting a chemically inert surface to the wound fluid and preventing non-specific adsorption of wound proteins to the bare gold.

Close-packed monolayer density

The maximum surface density of a fully close-packed alkanethiol monolayer on gold, measured by reductive desorption coulometry, is [6]:

$$\Gamma_{\max} = 9.3 \times 10^{-10} \text{ mol cm}^{-2} = 930 \text{ pmol cm}^{-2} \tag{I.1}$$

Love et al. report an independent cross-check of approximately 750 pmol cm^{-2} for densely packed alkanethiol monolayers on Au(111) [1]. Equation (I.1) is an upper bound. The probe-thiol cannot approach $\Gamma_{\max}$ due to the bulky peptide headgroup; the target probe coverage $\Gamma_p = 2$–5 pmol cm^{-2} is 0.2–0.5% of the close-packed limit.

The following assumptions underlie this section:

1. The probe-thiol adsorbs to gold through the thiol–gold bond; MCH backfilling fills the remaining gold surface.
2. $\Gamma_{\max} = 930$ pmol cm^{-2} is an upper bound on MCH packing; the actual MCH density is lower because probe-thiols occupy a fraction of the surface.

II. Deposition Protocol and Target Surface Density

State of available literature

The papers in `papers/layer_07/` that address deposition protocol and coverage (Herne 1997 [2], Steel 1998 [3], Petrovykh 2003 [4]) were obtained for a DNA-thiol (aptamer) probe. Their specific results do not transfer to the peptide-thiol probe used in this device for the following reasons:

- **Herne 1997 [2]**: The 1.0 M KH_2PO_4 buffer requirement and the log-time deposition kinetics were demonstrated specifically for HS-ssDNA. The need for high ionic strength arises from screening electrostatic repulsion between the 25 ionisable phosphate groups per DNA strand. A short peptide probe carries far fewer charged groups and is not expected to require the same ionic-strength buffer.
- **Steel 1998 [3]**: The upper functional coverage limit (4×10^{12} molecules cm^{-2}) was determined by hybridisation steric hindrance, which is specific to complementary-strand binding. The relevant steric limit for enzymatic cleavage is set by MMP-9 (92 kDa, diameter ≈ 5–6 nm) accessing individual probe molecules, which imposes a different and likely more stringent spacing requirement.
- **Petrovykh 2003 [4]**: Non-Langmuir (log-linear) adsorption kinetics were shown specifically for HS-ssDNA. Peptide-thiol adsorption may follow different kinetics; the Langmuir model may be a better approximation for short peptide probes without long-range electrostatic repulsion.

No paper currently in `papers/layer_07/` provides peptide-thiol SAM deposition conditions, coverage data, or functional limits for enzyme-cleavage sensors. The probe loading section is therefore **unconfirmed**. Table 1 lists the papers the team must obtain.

Information needed	**Why needed**	**Suggested source**
Peptide-thiol deposition protocol on Au	Incubation conditions, buffer, time	Peptide-SAM sensor paper specific to MMP-9 cleavage assay
Γ_p vs. deposition conditions	Quantitative coverage control	Same paper or SPR/QCM calibration
Functional upper limit on Γ_p for enzyme accessibility	MMP-9 must physically access each probe site	Cleavage-efficiency vs. coverage study
k_{cat}, K_m for MMP-9 cleaving device peptide	Required for Layer 0 signal calibration (Layer 0 Limitation 5)	Kinetics paper for specific peptide sequence

Table 1: Papers required to complete the probe-loading section.

Probe spacing constraint from enzyme size

Once Γ_p is established from the literature, the average probe spacing is:

$$\ell = (\Gamma_p \cdot N_A)^{-1/2} \tag{II.1}$$

For the sensor to function, MMP-9 (molecular weight 92 kDa, hydrodynamic radius ≈ 3–4 nm) must approach and bind each surface-immobilised peptide. This imposes the requirement $\ell \gtrsim 2 \times r_{\text{MMP-9}} \approx 6$–$8$ nm, corresponding to $\Gamma_p \lesssim 3$ pmol cm^{-2}. At higher probe densities, neighbouring probe molecules sterically block enzyme access, reducing the effective cleavage rate and underestimating free MMP-9 activity. This constraint is more stringent than the hybridisation limit in Steel 1998 and must be confirmed with a cleavage-efficiency vs. coverage measurement.

The following assumptions underlie this section:

1. Probe-thiol adsorbs to gold via the thiol–gold bond; MCH backfilling fills remaining surface sites.
2. A target of $\Gamma_p \lesssim 3$ pmol cm^{-2} is adopted on steric grounds (enzyme-access requirement); this value is not confirmed from any paper in `papers/layer_07/` and is flagged as Limitation 1.

III. SAM Stability over the Assay Window

Three mechanisms can degrade the SAM during the 60-minute assay: oxidative desorption by H_2O_2, thiol exchange with free thiols in the wound matrix, and electrochemical desorption by the SWV scan potential.

Oxidative desorption

Schoenfisch and Pemberton characterised alkanethiol SAM oxidation under ambient conditions [5]. Their central finding is that ozone (O_3), not O_2 or UV, is the primary oxidant in air, converting Au–SR to Au^0 + RSO_xH. Under ambient O_3 at 28–60 ppb, a C_{12} SAM on gold lost 36.9% of its total sulphur in 1 h (Table 5 of [5]).

In the present device, the oxidant is H_2O_2 at 10–100 μM in aqueous wound fluid; ozone is absent from the aqueous phase. The relative oxidising power of the two species is reflected in their standard reduction potentials:

$$E^\circ(O_3/O_2) = +2.07 \text{ V}, \qquad E^\circ(H_2O_2/H_2O) = +1.76 \text{ V} \tag{III.1}$$

The 310 mV thermodynamic advantage of O_3 corresponds to a rate advantage of several orders of magnitude for thiolate oxidation. The Schoenfisch data therefore provide a conservative upper bound: even under the harshest oxidative conditions tested (ambient O_3 in air), SAM loss is approximately 37% in 1 h for a C_{12} chain. Aqueous H_2O_2 at wound-fluid concentrations is a substantially weaker oxidant; oxidative SAM loss over the 60-minute assay is not expected to approach this bound.

No quantitative rate constant k_{ox} for H_2O_2 oxidation of thiol SAMs in solution is available in `papers/layer_07/`. The argument above is comparative only; see Limitation 2.

Thiol exchange

Schlenoff et al. investigated the exchange of adsorbed alkanethiols on gold by free thiol in solution [7]. The key finding is that the exchange rate shows "very little dependence of rate on thiol concentration," meaning the process is not bimolecular but involves a rate-limiting surface step. In hexane, the pseudo-first-order exchange rate constant is [7]:

$$k_1 = 1.7 \times 10^{-5}\ \text{s}^{-1} \quad \text{(hexane, 22°C)} \tag{III.2}$$

The fraction exchanged at time t is $f = 1 - \exp(-k_1 t)$. At $t = 3600$ s (60 min):

$$f = 1 - \exp(-1.7 \times 10^{-5} \times 3600) = 1 - e^{-0.061} \approx 5.9\% \tag{III.3}$$

Equation (III.3) is a strict upper bound: Schlenoff et al. further show that exchange in aqueous conditions is negligible, with essentially no exchange detected even after 125 h in water (Figures 3–6 of [7]). They conclude that desorption is "least pronounced in aqueous systems." Wound fluid is an aqueous matrix; thiol exchange does not contribute meaningfully to SAM loss over the assay window.

Electrochemical desorption

Reductive desorption of alkanethiol SAMs on gold occurs via:

$$\text{Au–SR} + e^- \longrightarrow \text{Au}^0 + \text{RS}^- \tag{III.4}$$

and is only possible when the electrode potential reaches or exceeds the desorption potential E_{des}. Widrig et al. measured E_{des} for a series of ω-alkanethiol SAMs on gold by cyclic voltammetry in 0.5 M KOH, finding a linear dependence on carbon chain length n [6]:

$$E_{\text{des}} = (-0.83 - 0.020\,n)\ \text{V vs Ag/AgCl} \quad \text{(0.5 M KOH)} \tag{III.5}$$

For MCH ($n = 6$):

$$E_{\text{des}} = -0.83 - 0.020 \times 6 = -0.83 - 0.12 = -0.95\ \text{V} \tag{III.6}$$

Love et al. report that desorption of alkanethiol SAMs from gold typically occurs in the range -1.0 ± 0.25 V vs Ag/AgCl [1], giving a conservative lower bound of -0.75 V. The SWV scan window runs from -0.65 V to $+0.1$ V (Layer 6). The margin between the scan minimum and the conservative lower bound is:

$$\Delta E = E_{\text{des,min}} - E_{\text{scan,min}} = -0.75\ \text{V} - (-0.65\ \text{V}) = -0.10\ \text{V} \tag{III.7}$$

The scan potential is 100 mV more positive than the most conservative desorption estimate; the Widrig value for MCH is 300 mV more negative than the scan limit. Electrochemical desorption does not occur during the assay.

The following assumptions underlie this section:

1. The oxidative stability argument uses the Schoenfisch O_3/air benchmark as an upper bound; no quantitative H_2O_2 rate constant is claimed (Limitation 2).
2. Equation (III.2) and the 5.9% bound apply in organic solvent; aqueous exchange is negligible per Schlenoff [7] (Limitation 3).
3. The Love range [1] is used as the primary reference for device-condition E_{des}; the Widrig formula (equation III.5) is supporting data measured specifically in alkaline electrolyte (Limitation 4).

IV. Ca^{2+} Artefact and Ratiometric Cancellation

Mechanism

Wound fluid contains ionised Ca^{2+} at approximately 1.0–1.3 mM (similar to serum). If the probe-thiol architecture includes a PEG spacer between the alkyl chain and the peptide, Ca^{2+} can coordinate the polyether oxygen atoms in a crown-ether-like interaction, contracting the spacer and decreasing the probe-to-gold bridge length by Δd. Electron transfer (ET) from MB to gold through the bridge follows a tunnelling mechanism, so a decrease in bridge length raises the apparent ET rate constant k_{ET} and increases the SWV peak current of *uncleaved* probes.

For the peptide cleavage sensor (signal-off: MB released on cleavage), this Ca^{2+}-induced increase in k_{ET} acts on the uncleaved probe population equally before (I_0) and after ($I(t)$) the assay. The measured quantity is:

$$\frac{\Delta I}{I_0} = \frac{I_0 - I(t)}{I_0} \tag{IV.1}$$

If Ca^{2+} multiplies k_{ET} by a factor R for all uncleaved probes identically, then $I_0 \to R\,I_0$ and $I(t) \to R\,(1-f)\,I_0$ where f is the fraction cleaved. Substituting:

$$\frac{\Delta I}{I_0} = \frac{R\,I_0 - R(1-f)\,I_0}{R\,I_0} = f \tag{IV.2}$$

The Ca^{2+} factor R cancels exactly in the signal-off ratio, provided both I_0 and $I(t)$ are measured in the same buffer (same Ca^{2+} concentration). The signal is inherently self-normalising against Ca^{2+}-induced changes in electron transfer rate, as long as the measurement protocol does not change the buffer between baseline and post-incubation reads.

Tunnelling decay constant
The ET rate constant through a molecular bridge of length d is:

$$k_{ET} = k_{ET}^{0}\,\exp(-\beta\,d) \tag{IV.3}$$

where β is the tunnelling decay constant. Smalley et al. measured β for ferrocene-terminated (Fc) alkanethiol chains tethered to gold in 1.0 M $HClO_4$, by fitting the chain-length dependence of the standard rate constant k° [8]:

$$\beta = 0.94 \pm 0.06\ \text{Å}^{-1} = (9.4 \pm 0.6) \times 10^{9}\ \text{m}^{-1} \tag{IV.4}$$

This measurement used ferrocene (Fc) as the redox probe, while the present device uses methylene blue (MB). The value of β is adopted here because β is a property of the alkyl bridge that electrons tunnel through, not of the terminal redox group. Smalley et al. note that $\beta \approx 1.0\ \text{Å}^{-1}$ is consistent across multiple redox probes studied through the same alkyl bridge [8]. The Fc-derived value therefore transfers directly to the MB system.

Sensitivity of k_{ET} to bridge contraction
If Ca^{2+} coordination contracts the PEG spacer by Δd, the fold increase in k_{ET} follows from equation (IV.3):

$$\frac{k_{ET}^{Ca}}{k_{ET}^{0}} = \exp(\beta\,\Delta d) \tag{IV.5}$$

For a representative contraction of $\Delta d = 0.2$ nm (one carbon–carbon bond displacement, used as a sensitivity bound since Δd is not measured for this architecture; see Limitation 5):

$$\begin{aligned} \frac{k_{ET}^{Ca}}{k_{ET}^{0}} &= \exp(9.4 \times 10^{9}\ \text{m}^{-1} \times 0.2 \times 10^{-9}\ \text{m}) \\ &= \exp(1.88) \approx 6.5 \end{aligned} \tag{IV.6}$$

Both the active and control electrodes carry the same probe-thiol/MCH SAM and are exposed simultaneously to the same wound fluid, so this fold increase applies identically to both.

Residual error from probe density change after cleavage
Equation (IV.2) assumes that Ca^{2+} affects uncleaved probes with the same k_{ET} before and after the assay. This holds exactly if the cleaved probe stub (the fragment remaining on the electrode after MB release) has negligible electron transfer compared with the intact probe. If the stub retains some residual MB (partial cleavage) or if cleavage changes the spacer geometry in a way that modifies the Ca^{2+} response of neighbouring probes, a second-order residual arises. The magnitude of this residual depends on the specific peptide architecture and is not quantifiable without a calibration measurement. It is expected to be small (second-order in f and in $\delta k_{ET,stub}$), but experimental confirmation that $\Delta I/I_0$ is Ca^{2+}-independent in the assay buffer is recommended.

Limitations of this model

1. **Probe loading protocol and Γ_p are unconfirmed: no peptide-thiol SAM paper is present in `papers/layer_07/`.** The three papers that previously governed this section (Herne 1997 [2], Steel 1998 [3], Petrovykh 2003 [4]) were obtained for a DNA-thiol probe and do not apply to the peptide-thiol system (see Section II). The deposition protocol, target Γ_p, and functional coverage limits are all unconfirmed. The papers listed in Table 1 must be obtained before Γ_p can be specified. Calibration by coulometric integration (Widrig [6]) is still applicable as a coverage measurement method independent of probe identity.

2. **No quantitative k_{ox} for H_2O_2 oxidation of thiol SAMs.** Schoenfisch and Pemberton [5] measure SAM oxidation by O_3 in air, not by H_2O_2 in solution. No paper in `papers/layer_07/` provides k_{ox} for aqueous H_2O_2. The oxidative stability argument is comparative only; experimental confirmation of SAM integrity after 60 min in wound-fluid H_2O_2 is required before the assay window can be certified.
3. **Thiol exchange rate confirmed in hexane only.** Schlenoff et al. [7] measure $k_1 = 1.7 \times 10^{-5}$ s^{-1} in hexane and demonstrate negligible aqueous exchange qualitatively; a quantitative rate constant for exchange by Cys or GSH in phosphate-buffered wound fluid at pH 7.4 is not available.
4. **E_{des} measured in 0.5 M KOH; wound-fluid conditions differ.** The Widrig formula (equation III.5) was determined at pH $\approx$ 13.7 [6]. At wound-fluid pH 7.4 the Nernst equation predicts that E_{des} shifts approximately $+(13.7 - 7.4) \times 0.059 \approx +0.37$ V toward positive values (because RS^- is protonated at pH 7.4), placing E_{des} near $-0.95 + 0.37 = -0.58$ V. This is only 70 mV positive of the scan minimum of -0.65 V, compared with the 300 mV margin in alkaline electrolyte. The Love range [1] (-1.0 ± 0.25 V, conditions unspecified) is used as the primary working reference, but experimental confirmation of E_{des} in the assay buffer is recommended before the scan window is finalised.
5. **Ca^{2+} self-cancellation in $\Delta I/I_0$ is conditional on the measurement protocol.** Equation (IV.2) shows that the Ca^{2+} artefact cancels in $\Delta I/I_0$ only if I_0 and $I(t)$ are measured in the same buffer (same Ca^{2+} concentration). If the baseline I_0 is measured before adding wound fluid (in calibration buffer without Ca^{2+}) and $I(t)$ is measured in wound fluid (with Ca^{2+}), the cancellation fails and a spurious signal arises. Protocol design must ensure that both measurements are in the same ionic environment. Whether a PEG spacer is present and its Ca^{2+} affinity are also unconfirmed for this probe architecture.

References

All papers in `electrochemical_device/papers/layer_07/`.

[1] Love JC, Estroff LA, Kriebel JK, Nuzzo RG, Whitesides GM (2005). Self-assembled monolayers of thiolates on metals as a form of nanotechnology. *Chem Rev* 105(4):1103–1169. **love2005.pdf** – Γ_{max} cross-check ($\approx$ 750 pmol cm^{-2}, p. 1108), $E_{des} = -1.0 \pm 0.25$ V (p. 1122) confirmed from text.

[2] Herne TM, Tarlov MJ (1997). Characterization of DNA probes immobilized on gold surfaces. *J Am Chem Soc* 119(38):8916–8920. **herne1997.pdf** – *DNA-thiol specific. KH_2PO_4 buffer requirement, log-time deposition kinetics, and MCH displacement of non-thiolated DNA all pertain to HS-ssDNA with 25 charged phosphate groups per strand. These results do not transfer to the peptide-thiol probe and are not used for deposition protocol in this layer.*

[3] Steel AB, Herne TM, Tarlov MJ (1998). Electrochemical quantitation of DNA immobilized on gold. *Anal Chem* 70(22):4670–4677. **steel1998.pdf** – *DNA-thiol specific. The functional coverage limit (4×10^{12} molecules cm^{-2}) was determined by hybridisation steric hindrance. Enzyme-cleavage steric limits for MMP-9 are different and not covered by this paper.*

[4] Petrovykh DY, Kimura-Suda H, Whitman LJ, Tarlov MJ (2003). Quantitative analysis and characterization of DNA immobilized on gold. *J Am Chem Soc* 125(17):5219–5226. **petrovykh2003.pdf** – *DNA-thiol specific. Log-linear adsorption kinetics and the non-Langmuir conclusion apply to HS-ssDNA. Peptide-thiol adsorption kinetics are not covered by this paper.*

[5] Schoenfisch MH, Pemberton JE (1998). Air stability of alkanethiol self-assembled monolayers on silver and gold surfaces. *J Am Chem Soc* 120(18):4502–4513. **schoenfisch1998.pdf** – O_3 as primary oxidant confirmed from text; C_{12} SAM 36.9% sulphur loss in 1 h confirmed from Table 5. Paper does not give k_{ox} for aqueous H_2O_2; value attributed to this paper in `sam_stability.py` is not confirmed and has been removed.

[6] Widrig CA, Chung C, Porter MD (1991). The electrochemical desorption of *n*-alkanethiol monolayers from polycrystalline Au and Ag electrodes. *J Electroanal Chem* 310(1–2):335–359. **widrig1991.pdf** – $\Gamma_{max} = 9.3 \times 10^{-10}$ mol cm^{-2} (p. 351) confirmed; E_{des} formula (equation III.5) confirmed from text. Measured in 0.5 M KOH; pH correction noted in Limitation 4.

[7] Schlenoff JB, Li M, Ly H (1995). Stability and self-exchange in alkanethiol monolayers. *J Am Chem Soc* 117(50):12528–12536. **schlenoff1995.pdf** – $k_1 = 1.7 \times 10^{-5}$ s^{-1} in hexane confirmed from Table 3; concentration-independence of exchange rate confirmed from text; negligible aqueous exchange confirmed from Figures 3–6. Bimolecular rate model in `sam_stability.py` is not supported by this paper and has been removed.

[8] Smalley JF, Feldberg SW, Chidsey CED, Linford MR, Newton MD, Liu Y-P (2003). The kinetics of electron transfer through ferrocene-terminated alkanethiol monolayers on gold. *J Am Chem Soc* 125(7):2004–2013. **smalley2003.pdf** – $\beta = 0.94 \pm 0.06$ Å^{-1} confirmed from text (p. 2009). Measured using ferrocene (Fc) redox probe; value transfers to methylene-blue system because β is a property of the alkyl bridge, confirmed from discussion in [8].

[9] Becka AM, Miller CJ (1993). Electrochemistry at ω-hydroxy thiol coated electrodes. 3. Voltage independence of the electron tunneling barrier and measurements of redox kinetics at large overpotentials. *J Phys Chem*

97(23):6233–6239. **becka1993.pdf** – provides context for tunnelling through alkanethiol SAMs; the $\beta = 0.94\ \text{Å}^{-1}$ value used in this layer is from Smalley 2003 [8], not Becka 1993.